Further Assumptions and Alternative Models

The following section presents further assumptions and outlines the alternative models considered in the project.

The walking time between any two adjacent stations is 15 seconds.

The two alternative models mentioned in the project description are outlined below.

Alternative 1: Adding pallets to the fourth system

This alternative increases the number of pallets in the fourth system. Because the system's current control logic was designed specifically for 40 pallets, it must be modified before any pallets can be added. This modification has a fixed cost of $17,000, which applies even if only a single pallet is added. Each additional pallet also carries an incremental cost of $3,000.

Alternative 2: Adding buffer conveyors for incoming parts

This alternative adds three new buffer conveyors at the incoming merge point, one for each part type. Available floor space limits the expansion to these three conveyors, each with a capacity of ten parts. Each of the first three lines sends its parts to its own buffer conveyor. New release logic then transfers the parts to the existing buffer conveyor, which holds up to 40 parts, in batches of the same part type (for example, batches of five). Since every batch contains a single part type, the automated operations in the final system need only one setup at the start of each batch. The total cost of this alternative is $56,000, based on a quote from a conveyor manufacturer, and covers equipment, installation and the development of the release logic.

NS306 Productions Company requests that during your analysis you evaluate and answer the following questions:

1. What is the cost of lost production for our current system?

2. What is the cost savings, if any, resulting from the implementation of the first proposed alternative? How many additional pallets should be added?

3. What is the cost savings, if any, resulting from the implementation of the second proposed alternative? Include a detailed description of your proposed release logic.

4. Should we implement both proposed modifications?

Please provide all cost comparisons in dollars per week. Your report should include a recommendation for the most cost-effective system. Your analysis may reveal that none of the alternative options should be considered.

If you have any other valid and logical alternative scenarios to propose—except the proposed scenarios—please implement and report on them as well. Such scenarios could contribute to your project score.

The Findings from Real System

The findings from real system are given below:

Performance Criteria	Value (Minutes)	ε
Total Time for A100	73.90	7.39
Waiting Time for A100	65.70	6.57
Total Time for A200	74.00	7.40
Waiting Time for A200	66.50	6.65
Total Time for A300	74.10	7.41
Waiting Time for A300	66.10	6.61